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ROTOR ASSEMBLY, METHOD FOR MANUFACTURING ROTOR ASSEMBLY, AND MOTOR INCLUDING ROTOR ASSEMBLY

Abstract

A rotor assembly includes: a shaft; a rotor core having a through-hole which is formed in an axial direction (A) and into which the shaft is inserted; a magnet inserted into an inner space of the rotor core; and a plate provided in close contact with one side of the rotor core in the axial direction (A). The plate includes a protrusion formed in a region, which faces a region between the magnet and the circumference of the rotor core, and protruding toward the rotor core.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is the continuation application of U.S. patent application Ser. No. 17/879,025 filed Aug. 2, 2022, which claims priority from and the benefit of Korean Patent Application No. 10-2021-0101558, filed on Aug. 2, 2021, the entire contents of each are hereby incorporated by reference.

TECHNICAL FIELD

[0002] Exemplary embodiments relate to a rotor assembly, a method for manufacturing the rotor assembly, and a motor including the rotor assembly.

BACKGROUND

[0003] For motors including rotors and stators, in order to ensure the safety of a rotor core provided in a rotor even in the high-speed range of a motor, it is necessary to secure rigidity even in a region of the rotor core to which stress is concentrated.

[0004] The region of the rotor core, to which the stress is concentrated, includes a section between a circumferential region of the rotor core and a region provided with the magnet, and is a region in which the thickness of the rotor core is relatively small (hereinafter, referred to as a 'small thickness region'). According to the related art, the durability of the rotor core against stress is improved by increasing the thickness of the small thickness region.

[0005] However, according to the related art, when the thickness of the small thickness region increases, the amount of magnetic flux that leaks during a process of driving a motor increases, and accordingly, the amount of effective magnetic flux decreases. This causes deterioration in torque and output of the motor.

SUMMARY

[0006] Exemplary embodiments of the present invention provide a motor and a rotor assembly, which have a new structure that can prevent deterioration in torque and output of the motor while resolving limitations caused by the concentration of stress in a specific region of the rotor core.

[0007] A first exemplary embodiment of the present invention provides a rotor assembly including: a shaft; a rotor core having a through-hole which is formed in an axial direction (A) and into which the shaft is inserted; a magnet inserted into an inner space of the rotor core; and a plate provided in close contact with at least one side of the rotor core in the axial direction (A), wherein the plate includes a protrusion formed in a region, which faces a region between the magnet and the circumference of the rotor core, and protruding toward the rotor core.

[0008] The protrusion may be provided in close contact with the magnet.

[0009] A certain region may be formed between the rotor core and the magnet, and the rotor assembly may further include a resin portion that is provided in the certain region and bonded to the rotor core and the magnet.

[0010] The protrusion may be provided in close contact with the resin portion.

[0011] The magnet may be provided inside the rotor core at a certain angle (θ) with respect to a radial direction (R) perpendicular to the axial direction (A), and the protrusion may be provided in close contact with one end of the magnet that is adjacent to the circumference of the rotor core.

[0012] The resin portion may include a first resin portion bonded to one end of the magnet that is adjacent to the circumference of the rotor core.

[0013] The resin portion may further include a second resin portion bonded to the other end of the magnet on the opposite side from the one end.

[0014] An empty space may be formed between the magnet and the rotor core at the other end of

the magnet on the opposite side from the one end.

[0015] The magnet may include: a first magnet provided at the certain angle (θ) with respect to the radial direction (R); and a second magnet located parallel to the first magnet and provided further inside the rotor core than the first magnet, wherein the protrusion includes: a first protrusion provided in close contact with one end of the first magnet that is adjacent to the circumference of the rotor core; and a second protrusion provided in close contact with one end of the second magnet that is adjacent to the circumference of the rotor core.

[0016] The plate may include: a first plate provided in close contact with one side end of the rotor core in the axial direction (A); and a second plate provided in close contact with the other side end of the rotor core in the axial direction (A).

[0017] A second exemplary embodiment of the present invention provides a motor including: a rotor assembly; and a stator provided on one side of the rotor assembly in a radial direction (R), wherein the rotor assembly includes: a shaft; a rotor core having a through-hole which is formed in an axial direction (A) and into which the shaft is inserted; a magnet inserted into an inner space of the rotor core; and a plate provided in close contact with at least one side of the rotor core in the axial direction (A), wherein the plate includes a protrusion formed in a region, which faces a region between the magnet and the circumference of the rotor core, and protruding toward the rotor core.

[0018] A third exemplary embodiment of the present invention provides a method for manufacturing a rotor assembly, the method including: preparation operation of preparing a shaft, a rotor core, a magnet, and a plate; assembly operation of i) inserting the shaft into a through-hole formed inside the rotor core, ii) inserting the magnet into an inner space formed inside the rotor core, and iii) bringing the plate into close contact with at least one side end of the rotor core in an axial direction (A); and filling operation of filling a space between the rotor core and the magnet with a resin material having flowability, wherein the plate is provided with a protrusion that protrudes in one direction, wherein during the assembly operation, the plate is disposed such that the protrusion faces the rotor core.

[0019] During the assembly operation, the protrusion may be provided in close contact with the at least one side of the magnet.

[0020] During the filling operation, the resin material may fill the space so as to come into close contact with the protrusion.

[0021] The resin material may be an epoxy resin.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0022] The accompanying drawings, which are included to provide a further understanding of the invention and are incorporated in and constitute a part of this specification, illustrate embodiments of the invention, and together with the description serve to explain the principles of the invention.

[0023] FIG. 1 is a first perspective view illustrating a structure of a rotor assembly according to the present disclosure.

[0024] FIG. 2 is a second perspective view illustrating a structure of a rotor assembly according to the present disclosure.

[0025] FIG. 3 is a view when a state, in which a rotor core, a magnet, and a plate are coupled to each other in a rotor assembly according to the present disclosure, is viewed in an axial direction.

[0026] FIG. 4 is an enlarged view illustrating part of FIG. 3.

[0027] FIG. 5 is an enlarged view illustrating a plate according to the present disclosure.

[0028] FIG. 6 is a perspective view illustrating a state in which a rotor core is removed from a rotor assembly according to the present disclosure.

[0029] FIG. 7 is an enlarged perspective view illustrating a coupling relationship between a magnet

and a plate in a rotor assembly according to the present disclosure.

[0030] FIG. **8** is an enlarged perspective view illustrating a coupling relationship between a magnet, a plate, and a resin portion in a rotor assembly according to the present disclosure.

DETAILED DESCRIPTION

[0031] Hereinafter, a rotor assembly, a motor, and a method for manufacturing a rotor assembly according to the present disclosure will be described with reference to the drawings.

Rotor Assembly

[0032] FIG. **1** is a first perspective view illustrating a structure of a rotor assembly according to the present disclosure, and FIG. **2** is a second perspective view illustrating a structure of a rotor assembly according to the present disclosure. FIG. **3** is a view when a state, in which a rotor core, a magnet, and a plate are coupled to each other in a rotor assembly according to the present disclosure, is viewed in an axial direction, and FIG. **4** is an enlarged view illustrating part of FIG. **3**. Also, FIG. **5** is an enlarged view illustrating a plate according to the present disclosure, and FIG. **6** is a perspective view illustrating a state in which a rotor core is removed from a rotor assembly according to the present disclosure. FIG. **7** is an enlarged perspective view illustrating a coupling relationship between a magnet and a plate in a rotor assembly according to the present disclosure, and FIG. **8** is an enlarged perspective view illustrating a coupling relationship between a magnet, a plate, and a resin portion in a rotor assembly according to the present disclosure.

[0033] Referring to the drawing, a rotor assembly **10** according to the present disclosure may include: a shaft **100**; a rotor core **200** having a through-hole H which extends in an axial direction A, wherein the shaft **100** is inserted into the through-hole H; a magnet **300** inserted into an inner space of the rotor core **200** which is formed separately from the through-hole H; and a plate **400** provided in close contact with the rotor core **200** in the axial direction A.

[0034] The shaft **100**, the rotor core **200**, the plate **400** may rotate together because fixed to each other. More preferably, the through-hole H may be formed in the central region of the rotor core **200** in a radial direction R, and the inner space, into which the magnet **300** is inserted, may be spaced apart from the through-hole H in the radial direction R.

[0035] Meanwhile, according to the present disclosure, the plate **400** may include a protrusion **430** formed in a region, which faces a region between the magnet **300** and a circumference **200a** of the rotor core **200**, and protruding toward the rotor core **200**. More specifically, referring to FIG. **7**, the protrusion **430** may be provided in close contact with the magnet **300**.

[0036] According to the related art, while the rotor assembly rotates, stress is concentrated on some regions of the rotor assembly. The stress described above is concentrated on a relatively thin region, which is a region between the magnet and the circumference of the rotor core.

[0037] In the present disclosure, the plate **400** is provided with the protrusion **430** to cope with the stress concentration described above, and the protrusion **430** supports the magnet **300**. Accordingly, some of the stress concentrated on the rotor core according to the related art may be distributed to the protrusion **430**. In this case, the rigidity of the rotor assembly **10** in the radial direction R may be increased without increasing the thickness of the region between the magnet **300** and the circumference **200a** of the rotor core **200** in the radial direction R.

[0038] Meanwhile, when the radial thickness of the region between the magnet and the circumference of the rotor core increases, the amount of leaked magnetic flux, among the amount of the magnetic fluxes formed by the magnet during the rotation of the rotor assembly, increases. Accordingly, the amount of effective magnetic flux decreases, and this causes deterioration in output and torque of the motor.

[0039] However, according to the present disclosure, the thickness of the region between the magnet **300** and the circumference **200a** of the rotor core **200** in the radial direction R may be minimized by the protrusion **430**, and thus, the reduction in amount of effective magnetic flux may be prevented. As a result, the output and torque of the motor may be maximized. Hereinafter, the structure of the rotor assembly **10** according to the present disclosure will be described with

reference to the drawings.

[0040] Referring to FIGS. 3 and 4, a certain region may be formed between the rotor core **200** and the magnet **300**. This may be understood as indicating that a space is formed between the rotor core **200** and the magnet **300**.

[0041] Here, according to the present disclosure, a resin portion **500** may be further included, which is provided in the certain region described above and bonded to each of the rotor core **200** and the magnet **300**. This may be understood as indicating that the space between the rotor core **200** and the magnet **300** is filled with the resin portion **500**. More preferably, the protrusion **430** may be provided in close contact with the resin portion **500**. Therefore, the protrusion **430** may support not only the magnet **300** but also the resin portion **500**. The resin portion **500** may be, for example, an epoxy resin, but the material of the resin portion **500** is not limited thereto.

[0042] Meanwhile, as illustrated in FIGS. 3 and 4, the magnet **300** may be provided inside the rotor core **200** at a certain angle θ with respect to the radial direction R perpendicular to the axial direction A. Also, referring to FIGS. 4 to 7, the protrusion **430** may be provided in close contact with one end of the magnet **300** that is adjacent to the circumference **200a** of the rotor core **200**.

[0043] Meanwhile, referring to FIGS. 3, 4, and 8, the resin portion **500** may include a first resin portion **510** bonded to one end of the magnet **300** that is adjacent to the circumference **200a** of the rotor core **200**. Also, the resin portion **500** may further include a second resin portion (not shown) bonded to the other end of the magnet **300** on the opposite side from the one end. However, the rotor assembly **10** according to the present disclosure may not include the second resin portion. In this case, an empty space, which is not filled with the resin portion **500**, may be formed between the magnet **300** and the rotor core **200** at the other end of the magnet **300** on the opposite side from the one end.

[0044] Meanwhile, the rotor core **200** may be provided in plurality. In one example, the rotor core **200** may include a first rotor core **210** and a second rotor core **220** provided in close contact with one side end of the first rotor core **210** in the axial direction A. However, unlike the above, the rotor core **200** may not be provided in plurality but may be provided as a single body, or may be divided into three or four sections. In the drawing, the rotor core **200** is illustrated as being divided into two sections as one example.

[0045] The plate **400** may be provided in close contact with each of both ends of the rotor core **200** in the axial direction A. More specifically, the plate **400** may include: a first plate **410** provided in close contact with one side end of the rotor core **200** in the axial direction A; and a second plate **420** provided in close contact with the other side end of the rotor core **200** in the axial direction A. As illustrated in the drawing, when the rotor core **200** includes the first rotor core **210** and the second rotor core **220**, the first plate **410** may be provided in close contact with one side end of the first rotor core **210** in the axial direction A, and the second plate **420** may be provided in close contact with one side end of the second rotor core **220** in the axial direction A. When one of the first plate **410** and the second plate **420** is a front rotor plate, the other may be a rear rotor plate.

[0046] Meanwhile, the magnet **300** may include: a first magnet **310** provided at the certain angle θ with respect to the radial direction R; and a second magnet **320** located parallel to the first magnet **310** and provided further inside the rotor core **200** than the first magnet **310**.

[0047] Also, the protrusion **430** may include: a first protrusion **431** provided in close contact with one end of the first magnet **310** that is adjacent to the circumference **200a** of the rotor core **200**; and a second protrusion **432** provided in close contact with one end of the second magnet **320** that is adjacent to the circumference **200a** of the rotor core **200**. Thus, the first protrusion **431** and the second protrusion **432** may be configured to support the first magnet **310** and the second magnet **320**, respectively.

[0048] Meanwhile, in one example, each of the first magnet **310** and the second magnet **320** may be provided in plurality. For example, as illustrated in FIGS. 3 and 4, two first magnets **310** may meet each other to form a V shape, and two second magnets **320** may meet each other to form a V

shape.

Motor

[0049] A motor according to the present disclosure may include: a rotor assembly **10**; and a stator provided on one side of the rotor assembly **10** in a radial direction R.

[0050] Here, the rotor assembly **10** may include: a shaft **100**; a rotor core **200** having a through-hole H which is formed in an axial direction A and into which the shaft **100** is inserted; a magnet **300** inserted into an inner space of the rotor core **200**; and a plate **400** provided in close contact with one side of the rotor core **200** in the axial direction A. Also, the plate **400** may include a protrusion **430** formed in a region, which faces a region between the magnet **300** and a circumference **200a** of the rotor core **200**, and protruding toward the rotor core **200**. The above description of the rotor assembly according to the present disclosure may be applied in the same manner to the motor according to the present disclosure, and thus, the details of the rotor assembly provided in the motor according to the present disclosure are replaced with the above description of the rotor assembly according to the present disclosure.

Method for Manufacturing Rotor Assembly

[0051] Referring to the drawings, a method for manufacturing a rotor assembly may include: a preparation operation of preparing a shaft **100**, a rotor core **200**, a magnet **300**, and a plate **400**; an assembly operation of i) inserting the shaft **100** into a through-hole H formed inside the rotor core **200**, ii) inserting the magnet **300** into an inner space formed inside the rotor core **200**, and iii) bringing the plate **400** into close contact with one side end of the rotor core **200** in an axial direction A; and a filling operation of filling a space between the rotor core **200** and the magnet **300** with a resin material having flowability.

[0052] Here, according to the present disclosure, the plate **400** may be provided with a protrusion **430** that protrudes in one direction. During the assembly operation, the plate **400** may be disposed such that the protrusion **430** faces the rotor core **200**. More specifically, during the assembly operation, the protrusion **430** may be provided in close contact with one side of the magnet **300**. More preferably, during the assembly operation, the protrusion **430** may be provided in close contact with one end of the magnet **300** that is adjacent to the circumference **200a** of the rotor core **200**.

[0053] Meanwhile, during the filling operation, the resin material may fill the space so as to come into close contact with the protrusion **430**, and the resin material may be an epoxy resin. The resin material may become the above-described resin portion **500** after being cured. Meanwhile, the above description of the rotor assembly according to the present disclosure may be applied in the same manner to the method for manufacturing a rotor assembly according to the present disclosure.

[0054] According to the present disclosure, provided are a motor and a rotor assembly, which have a new structure that can prevent deterioration in torque and output of the motor while resolving limitations caused by the concentration of stress in a specific region of the rotor core.

[0055] Although the present disclosure has been described with specific exemplary embodiments and drawings, the present disclosure is not limited thereto, and it is obvious that various changes and modifications may be made by a person skilled in the art to which the present disclosure pertains within the technical idea of the present disclosure and equivalent scope of the appended claims.

Claims

1. A rotor assembly comprising: a shaft; a rotor core having a through-hole extending in an axial direction (A), wherein the shaft is inserted into the through-hole; a magnet disposed in an inner space of the rotor core; and a plate disposed in contact with at least one side of the rotor core in the axial direction (A), wherein the plate has a first thickness and a second thickness greater than the first thickness, a region of the plate having the second thickness being disposed adjacent to the

magnet and a circumference of the rotor core.

2. The rotor assembly of claim 1, wherein the region of the plate having the second thickness is disposed between the magnet and the circumference of the rotor core and directly contacts the magnet.

3. The rotor assembly of claim 2, further comprising a resin portion extending in a direction parallel to the axial direction (A) and bonded to the rotor core and the magnet.

4. The rotor assembly of claim 3, wherein the region of the plate having the second thickness is in direct contact with the resin portion.

5. The rotor assembly of claim 2, wherein the magnet is disposed inside the rotor core and extends in a first direction parallel to the axial direction (A) and a second direction different from the first direction, the second direction having a certain angle (θ) with respect to a radial direction (R) perpendicular to the axial direction (A), and the region of the plate having the second thickness is disposed in contact with one end of the magnet in the second direction that is adjacent to the circumference of the rotor core.

6. The rotor assembly of claim 3, wherein the resin portion comprises a first resin portion bonded to one end of the magnet that is adjacent to the circumference of the rotor core.

7. The rotor assembly of claim 6, wherein the resin portion further comprises a second resin portion bonded to another other end of the magnet on an opposite side from the one end.

8. The rotor assembly of claim 6, wherein a space between the rotor core and another other end of the magnet on an opposite side from the one end is free of a resin portion.

9. The rotor assembly of claim 5, wherein the magnet comprises: a first magnet extending in the second direction having the certain angle (θ) with respect to the radial direction (R); and a second magnet extending parallelly to the first magnet and located further inside the rotor core than the first magnet, wherein the region of the plate having the second thickness comprises: a first region disposed in direct contact with one end of the first magnet that is adjacent to the circumference of the rotor core; and a second region disposed in direct contact with one end of the second magnet that is adjacent to the circumference of the rotor core.

10. The rotor assembly of claim 9, wherein the plate comprises: a first plate disposed in direct contact with one side end of the rotor core in the axial direction (A); and a second plate disposed in direct contact with the other side end of the rotor core in the axial direction (A).

11. The rotor assembly of claim 9, wherein the magnet further comprises: a third magnet extending in a third direction, having the certain angle (θ) with respect to the radial direction (R), the third magnet being formed symmetrically with the first magnet with respect to the radial direction (R) such that the first and third magnets form a 'V' shape; and a fourth magnet extending parallelly to the third magnet and located further inside the rotor core than the third magnet, the fourth magnet being formed symmetrically with the second magnet with respect to the radial direction (R) such that the second and fourth magnets form a 'V' shape.

12. The rotor assembly of claim 11, wherein a group of the first to fourth magnets is formed in plural along the circumference of the rotor core with an interval.

13. A motor comprising: the rotor assembly of claim 1; and a stator provided on one side of the rotor assembly in a radial direction (R).
